

Aditya Narendra

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EDUCATION

Moscow Institute of Physics & Technology
Bachelor's in Applied Mathematics & Computer Science

Moscow, Russia
Sept 2021 – July 2025

Mohamed Bin Zayed University of Artificial Intelligence
Master's in Robotics

Abu Dhabi, UAE
August 2026 – July 2028

EXPERIENCE

Research Engineer
MBZUAI - Mohamed bin Zayed University of Artificial Intelligence

January 2026 – August 2026
Abu Dhabi, UAE

- Worked on developing symbolic planner and controller for High-DOF Manipulators
- Validated the performance of the model on real Franka robot.
- Collaborated with researchers from MBZUAI, TU Munich and NYU-AD.

Research Engineer
Moscow Institute of Physics & Technology

December 2023 – January 2026
Moscow, Russia

- Contributed to RL and mobile manipulation research with publications (IROS, CoRL, etc.).
- Built IsaacSim environments for manipulators and multi-task RL benchmarking.
- Executed sim-to-real transfer and real-robot testing on the UR5 manipulator.

Teaching Assistant
Moscow Institute of Physics and Technology

Sept 2023 – Dec 2024
Moscow, Russia

- Supported programming courses for freshmen, seniors, and first-year master's cohorts.
- Mentored 40+ students in C++, Python, OOP, and DevOps through labs and tutorials.
- Evaluated assignments and exams and ensured consistent grading and feedback quality.

Full-Stack Developer Intern
Shioft

April 2023 – May 2023
Remote

- Developed a Django-based interface for an OpenCV-powered employee attendance system.
- Designed and implemented the backend database using MySQL.
- Worked on Computer Vision scripts and for integration with the web application.

PROJECTS

Model Predictive Control | *Python, C++, ROS, Casadi*

Dec 2023 – Feb 2024

- * Developed a ROS interface with Model Predictive Control (MPC) for a differential drive robot.
- * Integrated neural network-based cost functions and GPT models for obstacle avoidance.
- * Deployed and tested the code on a real Husky UGV robot.

Bi-Manual Loco-Manipulation Pipeline (Isaac-Sim) | *Isaac-Sim, IsaacLab, Python, RL*

Mar 2024 – Present

- * Built a bi-manual loco-manipulation training pipeline in Isaac-Sim/IsaacLab for whole-body control.
- * Implemented task suite and evaluation metrics across manipulation, locomotion, & loco-manipulation.
- * Benchmarked multi-task RL policies and enabled scalable parallel simulation for training.

Super Cuphead | *Python, PyGame, Multi-Agent RL*

2024

- * Built a Super Mario-style 2D platformer completely from scratch using Python and PyGame.
- * Implemented multi-agent RL to control enemy sprites/villains with adaptive behaviors.
- * Designed levels, collisions, health/score systems, and full gameplay loop with audio/FX.

PUBLICATIONS

CSymPlan: Certified Symbolic Planning & Control for High-DOF Manipulators <i>Under Review: International Journal of Research and Review (IJRR)</i>	Aug 2026
M4PO: Multi-Task Multi-Embodiment Model-Based Policy Optimization <i>The Conference on Robot Learning (CoRL'26)</i>	March 2026
Knowledge-Guided Manipulation Using Multi-Task Reinforcement Learning <i>IEEE International Conference on Robotics & Automation (ICRA'26)</i>	Sept 2025
Dynamic Neural Potential Field: Online Trajectory Optimization <i>IEEE International Conference on Robotics & Automation (ICRA'26)</i>	Sept 2025
Mind and Motion Aligned: A Joint Evaluation IsaacSim Benchmark <i>AI Journey Science Journal 2025</i>	Aug 2025
M3PO: Massively Multi-Task Model-Based Policy Optimization <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS'25)</i>	March 2025
Leveraging Multi-Task RL algorithms for Mobile Aloha Robot <i>International Conference on Intelligent Information Technologies for Industry (IITI'24)</i>	July 2024

ACHIEVEMENTS

- Winner of the 2025 Open Doors Scholarship, securing 5th place globally.
- Member of Foreign Professional Delegate at CIEP'24 Shanghai, China.
- Presented work on RL based controls for Mobile Aloha robot at DataFest 2024.
- Received the Rossotrudnichestvo scholarship from Ministry of Education, Russia.
- Serving as an Ambassador for Gamma.app

TECHNICAL SKILLS

Coursework: Machine Learning, Probability Theory, Convex Optimization, Advanced Combinatorics.
Languages: C/C++, Python, Assembly, SQL.
Frameworks: PyCharm, JAX, OpenAI Gym, StableBaselines3, Ray RLlib, HDFS, Spark.
Tools: Isaac Sim, IsaacLab, Mujoco, ROS, Gazebo.